

CMSC 471 Programming Assignment 1

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Question 1 Heuristics and admissibility

Let $M(s)$ be the positions where the current word s differs from the goal g . The heuristic estimates the cost that is still needed to reach g .

a. Steps

$$h(s) = |M(s)|$$

I count how many letters are in the wrong positions. Each of those positions has to change at least once, and one move only changes one position. Since each move costs 1, this count cannot be more than the actual remaining cost. For example, DOG and CAT differ in three positions, so $h(\text{DOG}) = 3$.

b. Scrabble

$$h(s) = \text{sum of value}(g[i]) \text{ for every } i \text{ in } M(s)$$

I add the Scrabble values of the goal letters that still need to be inserted. I use the values from the assignment, including 6 for J and X. Each wrong position must eventually receive its goal letter, so its insertion cost cannot be avoided. Extra changes only increase the cost. For DOG to CAT, the estimate is $\text{value}(C) + \text{value}(A) + \text{value}(T) = 3 + 1 + 1 = 5$.

c. Frequency

$$m(i,c) = \text{minimum rarity}(w) \text{ among dictionary words } w \text{ of the correct length with } w[i] = c.$$
$$h(s) = \text{sum of } [1 + m(i,g[i])] \text{ for every } i \text{ in } M(s).$$

For each wrong position, I find the smallest rarity of any legal word that has the goal letter in that position. I add 1 plus that minimum to the estimate. The minimum is available because the goal itself is in the dictionary.

Each wrong position must be fixed by a separate move. The word entered by that move has the required goal letter in that position, so its rarity is at least the minimum used in the formula. Adding these necessary costs cannot exceed the total cost of a solution. Some of the cheapest words may not be reachable at that point in a path, but using them can only make the estimate smaller.

All three heuristics return 0 at the goal because there are no wrong positions. They are nonzero away from the goal and depend on the selected cost measure.

Question 2 Does the heuristic help

Each search used a new DC instance. The table uses the expansion counts printed by the supplied search functions with display=True. Costs are rounded to five decimal places when needed.

Reduction (%) = $100 \times (\text{UCS expansions} - \text{A* expansions}) / \text{UCS expansions}$

Word pair	Cost	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction
WHY to ASK	Steps	5	773	5	16	97.9%
WHY to ASK	Scrabble	9	765	9	13	98.3%
WHY to ASK	Frequency	38.85541	808	38.85541	510	36.9%
BEAR to DUCK	Steps	4	691	4	6	99.1%
BEAR to DUCK	Scrabble	11	2981	11	6	99.8%
BEAR to DUCK	Frequency	38.24023	1084	38.24023	319	70.6%

1. Same optimal cost? Yes. UCS and A* returned the same optimal cost in all six comparisons. There were no cost discrepancies.
2. Largest reduction? Scrabble gave the largest reduction for both word pairs: 98.3% for WHY to ASK and 99.8% for BEAR to DUCK.
3. Least informative? Frequency gave the smallest reduction for both pairs: 36.9% for WHY to ASK and 70.6% for BEAR to DUCK.
4. Why are the results different? The steps heuristic counts needed letter changes, while Scrabble also accounts for expensive replacement letters. For BEAR to DUCK, these starting estimates are exactly the optimal costs: 4 steps and 11 Scrabble points. The frequency heuristic uses the cheapest rarity for each position separately. These words may not form a legal path, so the estimate can be much smaller than the real cost. Its starting estimates are about 10.19 and 11.35, compared with actual costs of 38.86 and 38.24. This leaves more competing paths for A* to explore.

Question 3 Reversibility and costs

Steps: Yes, any legal path can be reversed and keeps the same number of moves and reversing a shortest path proves that the optimal costs must be equal.

Scrabble: No. DOG to COG costs 3 because C is inserted. COG to DOG costs 2 because D is inserted. These direct moves are optimal: any path must eventually insert that goal letter in the first position and pay its value.

Frequency: No. DOG to COG costs $1 + 11.393166 = 12.393166$. COG to DOG costs $1 + 5.819885 = 6.819885$. These are dictionary rarities. Any path must pay the cost of entering its goal, so the direct move is optimal. Reversing an action is legal, but it can change which letter or word is charged.

Question 4 LLM heuristic audit

AI model: ChatGPT, identified as GPT-6 (Codex) in the original response.

Prompt:

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

AI proposal and formula: For each wrong position, find the minimum rarity of a legal word containing the goal letter in that position. Add 1 plus this minimum for every wrong position. Using the definitions from Question 1:

$hR(s) = \text{sum of } [1 + m(i, g[i])] \text{ for every } i \text{ in } M(s).$

Admissibility check: Consider any solution path. For each position that is wrong at the start, choose the first move that puts its goal letter there. These are separate moves because one move changes only one position. Each resulting word has rarity at least $m(i, g[i])$, so that move costs at least $1 + m(i, g[i])$. The heuristic only adds these necessary costs and leaves out other moves. Therefore, it cannot overestimate the cost of any solution.

Also, hR is at least h_0 because all rarities are nonnegative. It is strictly larger for at least one state: for DOG to COG, $h_0 = 1$ and $m(0, C) = 1.4666543$, so $hR = 2.4666543$. Both heuristics return 0 at the goal. The proposal is valid, so no repair is needed.

Experiment: A* was run with h_0 and hR on new frequency-cost problems. The verified hR is the frequency heuristic used in `dc.py`.

Word pair	h_0 cost	h_0 nodes expanded	hR cost	hR nodes expanded
WHY to ASK	38.85541	755	38.85541	510
BEAR to DUCK	38.24023	720	38.24023	319

Conclusion: Both heuristics gave the same optimal costs as UCS. hR used 245 fewer expansions for WHY to ASK and 401 fewer for BEAR to DUCK. These results support using hR for these problems. The proof establishes admissibility; the test results alone do not.

Sources and AI disclosure: ChatGPT/Codex helped to check and organize my thoughts and layout of the code and document. Verification included the supplied solver and test driver, input checks, and independent shortest-path checks. No outside code was downloaded.